

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER VII

Name of Department: - Computer Science and Engineering

1. Subject Code: **TCS 704** Course Title: **Advanced Computer Architecture**
2. Contact Hours: L: **3** T: **0** P: **0**
3. Examination Duration (Hrs): Theory **3** Practical **0**
4. Relative Weight: CIE **25** MSE **25** ESE **50**
5. Credits: **3**
6. Semester: **VII**
7. Category of Course: **DSC**
8. Pre-requisite: Logic Design and Computer Organization (TCS308)

9. Course Outcome:	After completion of the course the students will be able to: CO1 Discuss the classes of computers, and new trends and developments in computer architecture. CO2 Study advanced performance enhancement techniques such as pipelines, dynamic scheduling branch predictions, caches. CO3 Compare and contrast the modern computer architectures such as RISC, Scalar, and multi-CPU systems. CO4 Critically evaluate the performance of different CPU architecture. CO5 Improve the performance of applications running on different CPU architectures. CO6 Develop applications for high performance computing systems
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10. Details of the Course:

S. NO.	Contents	Contact Hours
1	Unit 1: Basics of Computing: Computer Architecture and Technology Trends, Moore's Law and its Applications, Classification of parallel computers, Performance based Computing, The Myopic View of Computer Architecture, Trends in Technology, Energy, Power and Cost, Dependability, Processor Speed, Cost, Power Consumption, Fabrication Yield	9

	Performance Metrics and Evaluation: Measuring Performance, Benchmark Standards, Amdahl's Law, Lhadma's Law	
2	Unit 2: Memory Hierarchy Design: Basics of Memory Hierarchy, Coherence and locality properties, Cache memory organizations, Advanced Optimization of Cache Performance, Memory Technology and Optimization, Cache Coherence and Synchronization Mechanism, Virtual Memory, Virtual Machines	10
3	Unit 3: Pipeline: Linear pipeline processor, nonlinear pipeline processor, Instruction pipeline Design, Dynamic instruction scheduling, Branch Handling techniques, branch prediction, Arithmetic Pipeline Design Data, and resource Dependences, Performance Issues in Pipeline: Pipeline Hazards, Data Hazards Branch hazards and Resource Hazards.	10
4	Unit 4: Instruction Level Parallelism: Concepts and Challenges, Basic Compiler techniques for exploiting ILP, Reducing the branch penalty with advanced branch predictions, overcoming data hazards with dynamic scheduling, exploiting ILP using multiple issues state scheduling	8
5	Unit 5: Multiprocessor architecture: Taxonomy of parallel architectures. Centralized shared-memory, distributed shared-memory architecture, Message passing vs Shared Memory Thread and Process Level Parallel Architecture: Instruction Level Data Parallel Architecture, SIMD Architecture, Fine Grained and Coarse-Grained Associative and Neural Architecture, Data Parallel Pipelined and Systolic Architectures, Vector Architectures	8
	Total	45

Textbooks:

Authors Name	Title	Edition	Publisher, Country	Year
Kai Hwang	Advance Computer Architecture	3 rd Edition	McGraw Hill	2017

Quinn	Parallel Computing: Theory & Practice	2 nd Edition	McGraw Hill	2017
Matthew	Beginning Linux Programming	2 nd Edition	SPD/WROX	2012
Hennessy and Patterson	Computer Architecture: A Quantitative Approach”,	1 st Edition	Elsevier	2011

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Quinn	Parallel Programming in C with MPI and Open MP	1 st Edition	McGraw Hill	2014